

The process through EDA of the data.

Purpose

This dataset explores how the use of artificial intelligence tools in education relates to students' academic performance, study habits, and learning outcomes.

Analytical questions

1. Study behavior

How does AI usage relate to the number of hours students spend studying?

2. Performance outcomes

Do students who use AI tools perform better academically than those who do not?

3. Understanding & learning

Is AI usage associated with better content understanding and learning outcomes?

Variables selected for analysis

- Study behavior analysis: AI usage indicators, AI usage time, and student study hours
- Academic performance analysis: AI usage purpose (e.g., exam preparation), grades, and final exam scores
- Learning and understanding analysis: Academic scores and performance improvement metrics

Initial dataset understanding steps

1. Identify the purpose of the data collection and the problem it aims to address
2. Review all columns to understand their meaning and potential relationships
3. Check for missing values and distinguish between missing data and non-applicable entries
4. Verify that each column uses the correct data type (e.g., numerical values, categorical fields, dates)

Column grouping for analysis

- **Student background:** age, gender, grade level
- **AI usage characteristics:** AI usage status, AI tools used, AI usage purpose, AI usage time, AI dependency level
- **Study behavior and habits:** study hours, attendance, sleep hours, learning focus areas
- **Academic performance:** subject scores, final exam score, grade, performance improvement indicators

Tooling and workflow choice

For this project, Google Sheets and Google Docs were used for data analysis and documentation. This choice was made to leverage Google Sheets' built-in version history, which allows tracking changes and recovering previous states without relying on Git version control. This approach ensures transparency, reproducibility, and safe iteration while focusing on analytical thinking rather than tooling complexity.

Key variables selected for initial analysis

- *Student background:* age, gender
- *AI usage and dependency:* AI usage time (minutes), AI dependency score
- *Study behavior and habits:* study hours per day, study consistency score, sleep hours
- *Learning and performance outcomes:* concept understanding score, final exam score, performance improvement rate

Session 1 Summary

In this session, the dataset was evaluated for suitability as a first data analysis project. Key analysis questions and objectives were defined, relevant variables were identified, and columns were grouped into logical analytical categories. A documentation and versioning workflow using Google Sheets was also established. No data cleaning or analysis was performed at this stage.

Session 2 – Missing Value Inspection (Preliminary)

Preliminary inspection indicates that key numerical variables (age, study hours per day, AI usage time) contain no significant missing values. Some missing entries appear in AI-related categorical fields and are likely structural, corresponding to students who do not use AI tools. Missing values in core behavioral and AI intensity variables (study hours per day, AI usage time, AI-generated content percentage) were identified as potentially high-risk for bias and will require careful handling during the cleaning phase.

Session 3 – Validity and Consistency Checks (Preliminary)

Value range and logic checks were performed on key variables including age, study hours per day, sleep hours, AI usage time, final score, and AI dependency score. Several values were reviewed for plausibility and internal consistency across related variables. No records were removed or modified at this stage; observations were documented to inform subsequent cleaning decisions. An inconsistency was observed between the binary `uses_ai` indicator and other AI-related variables. Some records marked as non-AI users (`uses_ai = 0`) still contain values for AI tools, AI usage time, or AI dependency. This suggests potential ambiguity in the definition of AI usage or survey design limitations. The `uses_ai` variable will therefore be treated cautiously and validated against other AI-related indicators during analysis.

Session 4 Data Consistency & AI Variables

In this session, the internal consistency of AI-related variables was examined before proceeding to analysis. Filtering rows where `uses_ai = 0` and `ai_tools_used = none` revealed that several students still had non-zero AI usage minutes, AI dependency scores ranging from 0 to 10, and notable AI-generated content percentages. Individual cases showed students with very low recorded AI usage time (e.g., 2 minutes) reporting high dependency scores and high AI-generated content, indicating that these variables do not measure the same construct.

These findings suggest that AI usage indicators in the dataset are perception-based and inconsistently defined, likely due to self-reporting and survey design limitations rather than data errors. As a result, no rows were removed or modified, and the dataset was deemed usable with caution. Subsequent analysis will prioritize continuous behavioral variables (such as AI usage time) and avoid strong causal claims while explicitly documenting these limitations.

Session 5 Documentation

In this session, basic exploratory analysis was conducted to compare high-performing students with the overall dataset. Students scoring 80 and above in the final score (353 students) were examined in terms of study hours and AI usage time. The average study hours (3.5) and AI usage minutes (89) for high scorers were nearly identical to the overall averages (3.3 study hours and 89 AI minutes), indicating that neither increased study time nor higher AI usage alone explains high academic performance in this dataset. This suggests that other factors beyond raw study hours and AI usage time contribute to higher scores.

Session 6 Documentation

In this session, learning outcomes were compared between high-performing students (final score ≥ 80) and the overall student population using concept understanding scores and improvement rates. High scorers showed a significantly higher average understanding score (8.3) compared to the overall average (5.5), while improvement rates were similar between the

two groups (10.6 vs 10). This indicates that stronger conceptual understanding, rather than short-term performance improvement, is more closely aligned with high academic performance in this dataset.

Session 7 Documentation

This session examined the relationship between AI usage time and conceptual understanding. Students were grouped into low AI usage (0–30 minutes) and high AI usage (90+ minutes) categories, and their average concept understanding scores were compared. The results showed nearly identical averages (5.3 for low usage and 5.5 for high usage), indicating no practically meaningful difference between the groups. This suggests that AI usage time alone does not appear to be associated with improved conceptual understanding in this dataset.

Session 8 Documentation

This session completed the final analytical question by examining the relationship between AI usage time and study hours. Students were grouped into low AI usage (0–30 minutes) and high AI usage (90+ minutes), and their average daily study hours were compared. Both groups showed the same average study time (3.3 hours), indicating no meaningful difference. This suggests that AI usage time does not appear to be associated with changes in how long students study in this dataset.

Session 9 Documentation — Final Conclusions

This project examined the relationship between AI usage, study behavior, learning outcomes, and academic performance among students. Across all analyses, AI usage time showed no meaningful association with study hours, conceptual understanding, or final academic performance. High-performing students did not study longer, use AI more, or show higher improvement rates than the overall population. Instead, conceptual understanding score emerged as the strongest factor aligned with high academic performance, suggesting that depth of understanding matters more than time spent studying or using AI tools.

The analysis also identified important dataset limitations. Several AI-related variables were inconsistent due to self-reported and perception-based measurements, particularly binary AI usage indicators and dependency scores. These limitations were documented rather than corrected, and conclusions were restricted to descriptive insights rather than causal claims. Overall, the dataset was deemed suitable for exploratory analysis with caution, and the project demonstrates a structured, transparent analytical workflow from problem definition through interpretation.

Final Answers to Analytical Questions

- **AI vs Study Hours:** No association observed

- **AI vs Performance:** AI usage time does not explain higher scores
- **AI vs Understanding:** AI usage time alone does not improve understanding
- **Key Insight:** Understanding quality is more important than AI usage quantity